

Chapter 11B: Inflammatory, Papulosquamous, Acneiform, and Allergic Skin Disorders

Medvale Internal Medicine - original study chapter for clinical learning and graph-RAG extraction

This chapter continues the dermatology section after the morphology overview. The supplied source segment covers acneiform eruptions, rosacea, hidradenitis suppurativa, atopic and contact dermatitis, psoriasis, seborrheic dermatitis, pityriasis rosea, lichen planus, urticaria, angioedema, anaphylaxis, and introductory drug reactions. The goal is practical recognition: morphology, distribution, time course, exposures, systemic danger, and treatment selection.

Learning objectives

- Separate acneiform, eczematous, papulosquamous, urticarial, angioedematous, anaphylactic, and drug-related eruptions using morphology, timing, distribution, and systemic features.
- Build first-line treatment pathways for acne, rosacea, hidradenitis suppurativa, atopic dermatitis, contact dermatitis, psoriasis, seborrheic dermatitis, pityriasis rosea, lichen planus, urticaria, angioedema, anaphylaxis, and fixed drug eruption.
- Recognize airway risk, severe cutaneous adverse reactions, erythroderma, infection mimics, and other red flags requiring urgent evaluation.

1. Pattern-based approach to common inflammatory rashes

Dermatologic diagnosis begins with description. Before naming the rash, identify the primary lesion and the pattern. Acneiform disease is follicular and often comedonal. Eczematous disease is inflamed, pruritic, poorly demarcated, and may ooze acutely or lichenify chronically. Papulosquamous disease forms papules or plaques with scale. Urticaria forms transient edematous wheals. Angioedema is deeper swelling of the skin or mucosa. Drug reactions often have a timing relationship to a new exposure and may range from benign exanthem to life-threatening severe cutaneous adverse reaction.

Distribution is often diagnostic. The central face suggests acne, rosacea, seborrheic dermatitis, perioral dermatitis, or lupus. Extensor surfaces and scalp suggest psoriasis. Flexural surfaces suggest atopic dermatitis. A sharply geometric or linear rash suggests contact dermatitis. Axillary and groin nodules with sinus tracts suggest hidradenitis. Migratory wheals suggest urticaria. Lip, tongue, or laryngeal swelling signals angioedema and airway risk. Diffuse erythema with fever, mucosal erosions, or skin pain is not a routine outpatient rash.

Pattern	Core morphology	Common distribution	Initial management frame
Acneiform	Comedones, follicular papules/pustules, nodules; rosacea lacks comedones.	Face, chest, back; HS in axilla/groin/folds.	Benzoyl peroxide/retinoid +/- antibiotics, hormonal therapy, isotretinoin, biologics or surgery by diagnosis/severity.
Eczematous	Ill-defined erythema, scale, vesicles/oozing acutely, lichenification chronically.	Atopic flexures; contact exposed/contact sites; seborrheic scalp/face/sebaceous areas.	Trigger avoidance, emollients, topical steroids, calcineurin inhibitors when needed.
Papulosquamous	Papules or plaques with scale.	Psoriasis extensor/scalp/nails; pityriasis trunk; lichen planus wrists/shins/oral/genital.	Topicals, phototherapy/systemic therapy for psoriasis; supportive care for pityriasis; steroids for lichen planus.
Urticarial/angioedema	Transient wheals or deeper nonpitting swelling.	Urticaria anywhere; angioedema eyelids/lips/tongue/hands/feet/GI.	Antihistamines for hives; airway assessment and epinephrine if anaphylaxis or laryngeal involvement.
Drug-related	Morbilloform rash, fixed dusky plaques, hives, pustules, bullae, mucosal erosions.	Often generalized; fixed eruption recurs at same site.	Stop culprit; supportive care for mild disease; urgent evaluation for SCAR features.

Red flags: dyspnea, wheeze, throat tightness, tongue/laryngeal swelling, hypotension, syncope, rapidly progressive rash, skin pain, duskiness, necrosis, extensive bullae, mucosal erosions, fever with toxicity, petechiae/purpura, erythroderma, facial edema with eosinophilia/hepatitis, ocular pain, or immunosuppression with widespread lesions.

2. Acne vulgaris

Acne vulgaris is a chronic inflammatory disorder of the pilosebaceous unit. Four processes interact: follicular hyperkeratinization plugs the follicular canal, androgen-responsive sebaceous glands increase sebum production, *Cutibacterium*

acnes proliferates in the obstructed follicle, and innate immune activation produces inflammatory papules, pustules, nodules, or cysts. The hallmark that distinguishes acne from many mimics is the comedone. Closed comedones appear as whiteheads; open comedones appear as blackheads because oxidized keratin and lipid darken at the follicular opening.

Severity is determined by lesion type, number, extent, scarring, and psychosocial burden. Mild comedonal acne has comedones and few inflammatory lesions. Moderate inflammatory acne has papules and pustules, often with trunk involvement. Severe nodulocystic acne has deep nodules, cysts, sinus tracts, scarring, or significant distress. Most topical regimens require at least 6 to 12 weeks of adherence before failure is declared unless irritation is intolerable.

Acne severity	Clinical picture	Preferred initial therapy	Escalation/cautions
Comedonal mild	Open/closed comedones with few inflammatory lesions.	Topical retinoid nightly as tolerated; add benzoyl peroxide if inflammatory component.	Start slowly if irritation; retinoids can irritate and are generally avoided in pregnancy.
Mild inflammatory	Papules/pustules with comedones, limited scarring risk.	Benzoyl peroxide + topical retinoid; topical clindamycin only with benzoyl peroxide.	Avoid topical antibiotic monotherapy; dryness and peeling are common early.
Moderate inflammatory	Numerous papules/pustules, trunk involvement, early scarring.	Topical combination therapy + oral doxycycline/minocycline/sarecycline if needed.	Limit oral antibiotic duration; maintain with topical benzoyl peroxide/retinoid.
Hormonal-pattern	Jawline/lower face, cyclic flares, adult female acne, hyperandrogen clues.	Combined oral contraceptive or spironolactone if appropriate.	Evaluate for PCOS/endocrine disease with virilization or irregular menses.
Severe nodulocystic/scarring	Nodules, cysts, sinus tracts, scarring, major psychosocial burden.	Dermatology referral for isotretinoin; bridge therapy as needed.	Isotretinoin is highly teratogenic; pregnancy prevention and monitoring are mandatory.

The 2024 American Academy of Dermatology acne guideline supports benzoyl peroxide, topical retinoids, topical antibiotics used appropriately in combination, oral doxycycline-class agents, hormonal therapies in appropriate patients, and isotretinoin for severe, scarring, psychosocially burdensome, or treatment-refractory acne. A practical acne regimen uses complementary mechanisms: open the follicle with a retinoid, reduce bacterial burden and resistance with benzoyl peroxide, suppress inflammation with short-term antibiotics when needed, and maintain remission after oral antibiotics with topical therapy.

Clinical pearl: vigorous scrubbing is unnecessary and may worsen acne by irritating the skin barrier. Gentle cleansing, noncomedogenic products, adherence, and time are more important than harsh washing. Diet is not the sole cause of acne, although high-glycemic patterns and skim milk can worsen acne in some patients.

Acneiform differential diagnosis

Acneiform eruptions without comedones should prompt reconsideration. Rosacea has centrofacial erythema, flushing, telangiectasia, papules/pustules, and ocular disease but no comedones. Folliculitis has follicle-centered pustules related to shaving, occlusion, bacteria, yeast, hot tubs, or immunosuppression. Perioral dermatitis clusters around the mouth and may worsen with topical steroids. Steroid acne causes monomorphic papules/pustules on trunk or face after systemic or potent topical steroid exposure. Acne fulminans is a severe systemic noduloulcerative variant with fever and arthralgias requiring urgent specialist care.

3. Rosacea

Rosacea is a chronic centrofacial inflammatory disorder, typically in adults, with recurrent flushing, persistent erythema, telangiectasia, inflammatory papules and pustules, ocular irritation, or phymatous thickening. Unlike acne vulgaris, rosacea lacks comedones. Common triggers include heat, alcohol, hot beverages, spicy foods, emotional stress, sunlight, and temperature changes. Patients may describe burning or stinging rather than simple itch.

Pattern	Findings	Treatment frame
Erythematotelangiectatic	Flushing, persistent central erythema, telangiectasia, burning/stinging.	Trigger avoidance, sunscreen, gentle skin care; brimonidine/oxymetazoline for erythema; laser/light for telangiectasia.
Papulopustular	Inflammatory papules/pustules on central face without comedones.	Topical metronidazole, azelaic acid, ivermectin; oral doxycycline for moderate-severe disease.
Phymatous	Thickened irregular skin, classically rhinophyma.	Dermatology referral; laser, electrosurgery, dermabrasion; isotretinoin sometimes used early.
Ocular	Blepharitis, conjunctival injection, foreign-body sensation, dry eye, chalazia.	Artificial tears/lid hygiene; doxycycline or ophthalmology evaluation if significant.

Rosacea is often mistaken for acne or lupus. Helpful discriminators are chronic flushing and telangiectasia for rosacea, comedones for acne, and photosensitive malar rash sparing nasolabial folds with systemic features for lupus. Chronic

topical steroid use on the face can worsen rosacea-like dermatitis and should be avoided unless a different diagnosis requires it.

4. Hidradenitis suppurativa

Hidradenitis suppurativa is a chronic, painful follicular occlusion disorder affecting intertriginous areas, especially axillae, groin, perineal/perianal skin, buttocks, and inframammary folds. Lesions begin as tender deep nodules or abscesses and evolve into draining sinus tracts, double-ended comedones, bridging scars, and chronic malodor. It is not simply recurrent poor-hygiene infection; the core process is follicular occlusion and immune dysregulation with secondary colonization. Smoking and obesity worsen activity, and onset commonly occurs after puberty.

Hurley stage	Clinical description	Management concepts
Stage I	Abscesses/nodules without sinus tracts or scarring.	Topical clindamycin, antiseptic washes, weight/smoking counseling, pain control, intralesional steroid for flares.
Stage II	Recurrent abscesses with sinus tracts and scarring, separated lesions.	Oral tetracycline-class antibiotics, selected clindamycin-rifampin regimens, hormonal therapy, dermatology referral.
Stage III	Diffuse involvement with interconnected sinus tracts and extensive scarring.	Biologic therapy such as adalimumab, multidisciplinary wound/pain care, surgical deroofting or excision.

Incision and drainage may temporarily relieve a tense abscess but does not control chronic hidradenitis and has high recurrence. Long-term care addresses inflammatory burden, sinus tracts, pain, psychosocial distress, smoking cessation, metabolic comorbidity, and scarring. Consider associated inflammatory bowel disease, spondyloarthritis, depression, and metabolic syndrome when symptoms suggest them.

5. Atopic dermatitis

Atopic dermatitis is a chronic relapsing eczematous disorder driven by skin barrier dysfunction, immune skewing, environmental triggers, and itch-scratch amplification. It often begins in childhood but may persist or start in adulthood. Patients may have personal or family history of atopy, including allergic rhinitis and asthma. Distribution varies by age: infants often have face/scalp/extensor involvement, while children and adults commonly have flexural dermatitis involving antecubital fossae, popliteal fossae, wrists, neck, eyelids, and hands.

Acute lesions are erythematous, edematous, vesicular, and weeping. Subacute lesions are scaly erythematous plaques. Chronic lesions become thickened, excoriated, and lichenified. Itch is central and can dominate sleep, school, work, and mood. Secondary infection should be suspected when eczema becomes honey-crusted, painful, pustular, rapidly worsening, or associated with fever. Eczema herpeticum is an emergency: monomorphic punched-out vesicles/erosions, pain, fever, and rapid spread in a patient with atopic dermatitis should prompt systemic antiviral therapy and urgent evaluation.

Treatment layer	Practical details	Common pitfalls
Barrier repair	Thick fragrance-free emollients, especially after bathing; gentle cleansers; avoid harsh soaps.	Underusing moisturizers; applying steroid without repairing barrier; using fragranced products.
Trigger reduction	Identify irritants/allergens, sweat/heat, wool, wet work, occupational exposures, stress, infection.	Over-restricting diet without evidence; missing occupational hand dermatitis.
Topical anti-inflammatory	Low potency steroids for face/folds; medium-high potency for trunk/extremities/lichenified plaques for limited courses; calcineurin inhibitors for eyelids/face/folds or maintenance.	Chronic high-potency steroid on thin skin; stopping without maintenance plan; undertreating flares.
Itch/infection strategy	Treat bacterial superinfection or eczema herpeticum when present; sedating antihistamines only for sleep in selected cases.	Relying on antihistamines as primary therapy; missing HSV dissemination.
Moderate-severe disease	Phototherapy, biologics, JAK inhibitors, or systemic immunomodulators under specialist care.	Repeated oral steroid bursts without steroid-sparing strategy.

Topical steroid potency is chosen by site, age, severity, and duration. Thin skin such as face, eyelids, groin, and intertriginous areas absorbs more medication and is more prone to atrophy, striae, telangiectasia, and periorificial dermatitis. Thick lichenified plaques on hands or feet often need stronger therapy for a short course. Maintenance may use intermittent anti-inflammatory therapy to recurrent sites plus daily barrier care.

6. Contact dermatitis

Contact dermatitis is inflammation caused by direct skin exposure. Irritant contact dermatitis is more common and results from direct chemical or physical injury to the barrier; prior sensitization is not required. Frequent handwashing, detergents, solvents, acids, alkalis, occlusion, and wet work are common causes. Allergic contact dermatitis is a type IV delayed

hypersensitivity reaction. Sensitization occurs after prior exposure, and rash appears after re-exposure, often 24 to 72 hours later, though the clinical window may be broader.

Feature	Irritant contact dermatitis	Allergic contact dermatitis
Mechanism	Direct barrier injury and innate inflammation.	Delayed type IV T-cell mediated hypersensitivity.
Prior exposure needed?	No.	Yes; sensitization precedes reaction.
Symptoms	Burning, stinging, pain, dryness/fissuring often prominent.	Pruritus prominent; vesicles/oozing in acute stage.
Borders/distribution	Limited to direct irritant exposure; hands common.	Geometric, linear, or patterned; may spread beyond contact area.
Common triggers	Soaps, detergents, solvents, acids/alkalis, wet work, friction.	Nickel, poison ivy/oak/sumac, fragrances, preservatives, rubber, neomycin, cosmetics.
Diagnosis	History and exam; occupational/repetitive exposure clues.	History/exam; patch testing if unclear, recurrent, occupational, or refractory.
Treatment	Avoid irritant, barrier repair, gloves/work modification, topical steroids.	Avoid allergen, topical steroids/cool compresses; systemic steroids for severe poison ivy.

Patch testing: evaluates delayed allergic contact dermatitis and is indicated when the allergen is not obvious, dermatitis is chronic/recurrent, occupational dermatitis is suspected, or the rash fails appropriate therapy. It is different from prick testing for immediate allergy.

7. Psoriasis

Psoriasis is a chronic immune-mediated disease characterized by accelerated keratinocyte proliferation and cutaneous inflammation. The classic form is chronic plaque psoriasis: well-demarcated erythematous plaques with thick silvery scale, often on extensor elbows/knees, scalp, umbilicus, lumbosacral area, palms/soles, and nails. Auspitz sign refers to pinpoint bleeding after scale removal. Koebner phenomenon means trauma can induce lesions in previously uninvolved skin. Nail pitting, onycholysis, oil spots, and subungual hyperkeratosis are important because nail disease correlates with psoriatic arthritis risk.

Psoriasis is more than a rash. Screen for joint pain, morning stiffness, dactylitis, heel pain/enthesitis, inflammatory back pain, uveitis, depression, metabolic syndrome, obesity, diabetes, hypertension, dyslipidemia, and cardiovascular risk. Genital, face, palm/sole, nail, scalp, or intertriginous psoriasis can be disabling even with limited body surface area.

Variant	Clues	Clinical concern
Plaque psoriasis	Well-demarcated erythematous plaques with thick scale on extensor/scalp/lumbosacral sites.	Chronic relapsing course; assess BSA, nails, joints, comorbidities.
Guttate psoriasis	Drop-like scaly papules often after streptococcal infection.	May resolve or evolve into chronic plaque disease.
Inverse psoriasis	Smooth erythematous plaques in folds with minimal scale.	Mimics intertrigo/candidiasis; lower-potency therapy needed in folds.
Pustular psoriasis	Sterile pustules; localized palms/soles or generalized with systemic illness.	Generalized pustular psoriasis can be severe and requires urgent care.
Erythrodermic psoriasis	Diffuse erythema and scaling involving most skin surface.	Thermoregulation, fluid loss, infection risk, high-output cardiac strain; urgent evaluation.
Nail psoriasis	Pitting, onycholysis, oil spots, subungual hyperkeratosis.	Strong clue to psoriatic arthritis risk.

Severity/site	Treatment approach	Comments
Mild plaque disease	Topical corticosteroids; vitamin D analogs; topical retinoids or calcineurin inhibitors for selected sites.	Combination steroid + vitamin D analog improves efficacy and steroid-sparing maintenance.
Face/folds/genitals	Low-potency steroid short course; calcineurin inhibitor or vitamin D analog carefully selected.	Avoid prolonged high-potency steroids due to atrophy and striae.
Scalp	Steroid solutions/foams/shampoos, keratolytics, vitamin D analogs.	Scale removal improves medication penetration.
Moderate-severe plaque or disabling special-site disease	Phototherapy, methotrexate, cyclosporine short rescue, acitretin, apremilast, biologics targeting TNF/IL-17/IL-23.	Choice depends on pregnancy, liver/kidney disease, infection risk, PsA, IBD, monitoring.
Psoriatic arthritis suspected	Rheumatology/dermatology coordination; systemic therapy preferred over skin-only topical treatment.	Ask about morning stiffness, swollen joints, dactylitis, enthesitis, inflammatory back pain.

Psoriasis steroid warning: avoid routine systemic corticosteroids for psoriasis when possible because withdrawal can precipitate severe flares or erythroderma. If systemic steroids are required for another life-threatening indication, psoriasis should be monitored and a steroid-sparing plan considered.

8. Seborrheic dermatitis

Seborrheic dermatitis is a chronic relapsing inflammatory dermatitis of sebaceous areas, associated with *Malassezia* yeast, host inflammatory response, oily skin, stress, fatigue, neurologic disease, and immunosuppression. It appears

as greasy yellow-white scale with mild erythema on the scalp, hairline, eyebrows, glabella, nasolabial folds, external ear canals, retroauricular areas, chest, axillae, and groin. In infants, cradle cap is a related presentation.

Treatment combines antifungal and anti-inflammatory strategies. Antifungal shampoos or creams such as ketoconazole, selenium sulfide, zinc pyrithione, or ciclopirox reduce *Malassezia* burden. Low-potency topical steroids or topical calcineurin inhibitors can be used briefly for inflamed facial/fold lesions. Because seborrheic dermatitis is chronic, maintenance therapy is often needed. Severe or refractory seborrheic dermatitis should raise consideration of HIV or other immunosuppressive states in the right context.

9. Pityriasis rosea

Pityriasis rosea is a self-limited papulosquamous eruption, classically beginning with a single herald patch that resembles tinea corporis: an oval scaly plaque, often on the trunk. Days to weeks later, multiple smaller oval plaques follow cleavage lines on the trunk, creating a Christmas tree distribution. Pruritus varies. It is not considered highly contagious and is thought to be associated with viral reactivation, often HHV-6/7.

Question	Pityriasis rosea answer
Who gets it?	Often adolescents and young adults, but it can occur across ages.
Where?	Trunk and proximal extremities; face is usually spared.
How long?	Usually resolves spontaneously in 6 to 8 weeks; recurrences are uncommon.
Main treatment?	Reassurance, antihistamines for itch, topical steroids for symptomatic lesions.
When to reconsider?	Palms/soles, mucosal lesions, systemic symptoms, high-risk sexual exposure, pregnancy, persistent/progressive rash, or absent herald patch with atypical distribution. Consider secondary syphilis, tinea, drug eruption, psoriasis, or viral exanthem.

10. Lichen planus

Lichen planus is a chronic inflammatory mucocutaneous disorder classically described by the 4 Ps: pruritic, purple, polygonal, planar papules/plaques. Common sites include wrists, ankles, shins, oral mucosa, genital mucosa, nails, and scalp. Oral lesions may show lacy white Wickham striae. Disease may be idiopathic or associated with hepatitis C, medications, dental materials, or other immune triggers. Hypertrophic lichen planus on shins can resemble chronic prurigo or psoriasis; erosive oral/genital disease can be painful and scarring.

Treatment depends on severity and site. High-potency topical corticosteroids are typical first-line therapy for cutaneous disease. Oral lesions may require topical steroid gels/rinses, calcineurin inhibitors, or specialist management. Widespread, hypertrophic, erosive, nail, scalp, or refractory disease merits dermatology referral. Biopsy is appropriate when the diagnosis is uncertain, lesions are atypical, mucosal erosions persist, or malignancy is a concern.

11. Urticaria

Urticaria consists of transient, pruritic, edematous wheals caused by mast cell mediator release and increased vascular permeability. Individual lesions usually appear and disappear within 24 hours, often migrating to new sites. They blanch with pressure and may have pale centers with erythematous borders. Acute urticaria lasts less than 6 weeks; chronic spontaneous urticaria lasts 6 weeks or longer and is often idiopathic or autoimmune rather than due to an easily identified external allergen.

Urticaria type	Clues	Treatment approach
Acute allergic/infection-associated	Sudden hives after food, medication, sting, viral illness; may include angioedema.	Remove trigger if known; second-generation H1 antihistamine; evaluate for anaphylaxis if respiratory, cardiovascular, GI, or mucosal involvement.
Physical inducible	Cold, heat, pressure, vibration, cholinergic/exercise, sunlight, water, dermatographism.	Avoid/modify trigger; prophylactic nonsedating antihistamines; cold urticaria requires swimming/anaphylaxis counseling.
Chronic spontaneous	Recurrent hives most days for more than 6 weeks, often no clear trigger.	Second-generation H1 antihistamines, dose escalation under guidance; omalizumab/specialist therapies if refractory.
Urticarial vasculitis mimic	Lesions last more than 24 h, painful/burning, leave bruising/hyperpigmentation, systemic symptoms.	Biopsy and systemic evaluation; do not manage as simple hives.

Urticaria discriminator: simple hives migrate and resolve without residual marks. Target lesions of erythema multiforme are fixed and have concentric zones. Urticarial vasculitis lesions are often fixed longer than 24 hours, painful or burning, and leave purpura or hyperpigmentation.

12. Angioedema

Angioedema is deeper swelling of dermis, subcutaneous tissue, or submucosa. It is often nonpitting, asymmetric, and more painful or pressure-like than itchy. Common sites are eyelids, lips, tongue, face, hands, feet, genitals, and GI tract. Angioedema can occur with urticaria in mast-cell mediated allergic reactions, or without urticaria in bradykinin-mediated disease such as ACE inhibitor-associated angioedema or hereditary/acquired C1 inhibitor deficiency.

Feature	Histamine/mast-cell angioedema	Bradykinin-mediated angioedema
Associated findings	Urticaria, pruritus, flushing, bronchospasm, hypotension possible.	No hives or itch; recurrent swelling, abdominal pain; airway risk.
Triggers	Foods, stings, drugs, latex, infection, idiopathic mast-cell activation.	ACE inhibitors, hereditary or acquired C1 inhibitor deficiency.
Response to antihistamines/epinephrine/steroids	Often responds, especially when allergic.	Often poor response; airway planning is still first priority.
Diagnostic clues	Temporal allergen exposure; tryptase may support anaphylaxis if timed.	Low C4 suggests C1 inhibitor deficiency; ACE inhibitor can cause swelling even after years.
Treatment priorities	Airway assessment; epinephrine if anaphylaxis; H1 antihistamines +/- steroids.	Airway assessment; stop ACE inhibitor; C1 inhibitor concentrate, icatibant, ecallantide, or FFP depending setting.

Airway rule: any tongue, floor-of-mouth, voice, stridor, drooling, progressive throat, or laryngeal symptom is an airway problem until proven otherwise. Medication treatment should never delay airway planning.

13. Anaphylaxis

Anaphylaxis is an acute systemic, potentially fatal hypersensitivity reaction. Skin and mucosal findings such as urticaria, flushing, pruritus, lip/tongue swelling, and periorbital edema are common and often early, but anaphylaxis can occur without skin findings. Respiratory findings include wheeze, dyspnea, hypoxemia, throat tightness, stridor, or airway obstruction. Cardiovascular findings include hypotension, syncope, tachycardia, shock, or collapse. GI findings include cramping, vomiting, and diarrhea. Common triggers include foods, medications, insect stings, latex, biologic agents, contrast agents, and exercise/cofactors in susceptible patients.

Step	Action	Rationale
1	Call for help; assess airway, breathing, circulation, mental status, skin/mucosa; remove trigger if possible.	Anaphylaxis can progress rapidly and recur biphasically.
2	Give intramuscular epinephrine immediately into mid-anterolateral thigh; repeat every 5-15 minutes if needed per protocol.	Epinephrine is first-line and treats airway edema, bronchospasm, vasodilation, and shock.
3	Place supine with legs elevated unless respiratory distress/vomiting requires another position; give high-flow oxygen.	Improves venous return and oxygen delivery.
4	Establish IV access; give isotonic fluids for hypotension; prepare airway support/intubation if progressive edema or respiratory failure.	Shock may require large-volume resuscitation.
5	Adjuncts: inhaled albuterol for bronchospasm; H1 antihistamines for hives/itch; H2 blockers and steroids may be used but do not replace epinephrine.	Adjuncts do not reliably reverse life-threatening airway or cardiovascular compromise.
6	Observe after stabilization; prescribe/teach epinephrine autoinjector and trigger avoidance; arrange allergy follow-up when appropriate.	Risk of biphasic reaction and recurrent exposure.

Epinephrine first: epinephrine delay is a major cause of preventable death in anaphylaxis. Antihistamines can improve itching and hives but do not treat shock, airway edema, or bronchospasm adequately. Corticosteroids have delayed onset and are not emergency rescue therapy.

14. Drug eruptions and fixed drug eruption

Adverse drug reactions are not always allergic. They include predictable pharmacologic adverse effects, interactions, toxicity, intolerance, pseudoallergic reactions, and immune-mediated hypersensitivity. A medication history should include prescriptions, over-the-counter drugs, supplements, intermittent analgesics, antibiotics, antiseizure medications, allopurinol, biologics, contrast exposure, and recent dose changes. Most simple morbilliform drug eruptions appear within days to weeks of starting a medication, but timing varies by prior sensitization and reaction type.

A fixed drug eruption is a sharply demarcated erythematous or violaceous patch/plaque that recurs at the same site with re-exposure to the culprit medication. It may blister and often leaves post-inflammatory hyperpigmentation. Common culprits include NSAIDs, antibiotics, acetaminophen, barbiturates, and other medications depending on population. Because it recurs in the same location, a careful history of prior identical lesions is diagnostically powerful.

Drug eruption pattern	Timing/morphology	Management
Morbilloform exanthem	Symmetric erythematous macules/papules, often trunk then limbs; usually days-weeks after drug.	Stop culprit if possible; topical steroids/antihistamines for symptoms; monitor for red flags.
Urticaria/angioedema	Hives and/or swelling minutes-days after exposure; may be IgE or non-IgE.	Assess for anaphylaxis; stop trigger; antihistamines or epinephrine if systemic reaction.
Fixed drug eruption	Recurrent sharply demarcated dusky plaque at same site; may blister; residual hyperpigmentation.	Identify and avoid culprit; topical steroids for symptoms.
AGEP	Acute widespread tiny sterile pustules with fever/neutrophilia, often within days.	Stop culprit; evaluate severity; supportive care/dermatology.
DRESS	Fever, facial edema, morbilliform rash, eosinophilia/atypical lymphocytes, hepatitis/nephritis/pneumonitis; often 2-8 weeks.	Stop culprit; urgent systemic evaluation; specialist care.
SJS/TEN	Skin pain, dusky macules, bullae, epidermal detachment, mucosal erosions, systemic toxicity.	Medical emergency; stop culprit; burn/ICU/dermatology/ophthalmology level care.

Drug rash red flags: drug rash plus fever, facial edema, mucosal involvement, skin pain, blistering, purpura, lymphadenopathy, eosinophilia, hepatitis, renal injury, dyspnea, or hypotension is not a simple outpatient exanthem.

15. Integrated diagnostic algorithms

Acneiform facial eruption

Question	If yes	If no
Are comedones present?	Acne vulgaris likely. Grade severity and treat with topical retinoid/benzoyl peroxide +/- antibiotics/hormonal/isotretinoin pathway.	Consider rosacea, folliculitis, perioral dermatitis, steroid acne, drug eruption, lupus, or demodex.
Flushing, telangiectasia, burning, ocular irritation, or rhinophyma?	Rosacea likely. Avoid triggers, sunscreen, topical metronidazole/azelaic acid/ivermectin, doxycycline if inflammatory.	Look for follicular pustules, monomorphic lesions, steroid exposure, or infection.
Painful nodules/abscesses in axilla/groin/folds with scars or sinus tracts?	Hidradenitis suppurativa. Stage disease; topical clindamycin to biologic/surgical pathway depending severity.	Return to acne/folliculitis/other differential.

Itchy eczematous eruption

Step	Interpretation
Flexural chronic relapsing itch + personal/family atopy.	Atopic dermatitis. Treat barrier and inflammation; watch for bacterial infection or eczema herpeticum.
Geometric/linear or exposure-pattern rash.	Contact dermatitis. Ask about work, hobbies, cosmetics, plants, adhesives, gloves, metals, topical medications; consider patch testing.
Greasy scale on scalp, eyebrows, nasolabial folds, ears, chest.	Seborrheic dermatitis. Antifungal shampoo/cream +/- brief low-potency anti-inflammatory therapy.
Annular scaly plaque with central clearing or asymmetric foot/groin involvement.	Consider dermatophyte infection before steroid escalation; KOH if uncertain.
Widespread erythroderma, fever, skin pain, mucosal lesions, immunosuppression.	Urgent evaluation. Do not assume routine eczema.

Papulosquamous eruption

Finding	Most likely diagnosis and next step
Thick silvery scale on extensor surfaces/scalp + nail pitting.	Psoriasis. Ask about psoriatic arthritis, metabolic disease, and special-site involvement.
Single herald patch then trunk oval plaques along cleavage lines.	Pityriasis rosea. Reassure/support unless atypical, pregnant, persistent, or STI risk.
Purple polygonal pruritic flat-topped papules on wrists/shins + oral lacy lesions.	Lichen planus. Consider hepatitis C/drug triggers; topical steroids; biopsy if atypical.
Greasy scale in sebaceous areas.	Seborrheic dermatitis. Treat Malassezia/inflammation.
Palms/soles, mucosa, systemic symptoms, sexual risk.	Consider secondary syphilis or systemic disease; test rather than assuming benign rash.

Hives, swelling, or allergic reaction

Question	Action
Any respiratory compromise, throat/tongue swelling, hypotension, syncope, severe GI symptoms, or multi-system involvement?	Treat as anaphylaxis: IM epinephrine first, ABCs, oxygen, IV fluids, airway preparation, observation.

Swelling without hives or itch, especially on ACE inhibitor or recurrent abdominal pain?	Consider bradykinin angioedema. Airway first; stop ACE inhibitor; consider C4/C1 inhibitor workup or targeted therapy.
Individual wheals resolve within 24 hours and migrate?	Urticaria. Use second-generation H1 antihistamines; identify triggers if acute; limited testing unless chronic or red flags.
Lesions last longer than 24 hours, bruise, burn/pain, systemic symptoms?	Consider urticarial vasculitis. Biopsy/labs and systemic evaluation.

16. High-yield comparison table

Disease	Signature clue	Common mistake	Treatment anchor
Acne vulgaris	Comedones plus inflammatory lesions.	Antibiotic monotherapy or stopping topicals too soon.	Benzoyl peroxide + topical retinoid; escalate by severity.
Rosacea	Flushing/telangiectasia without comedones.	Treating as acne or using chronic face steroids.	Trigger avoidance, sun-screen, metronidazole/azelaic acid/ivermectin.
Hidradenitis	Intertriginous painful nodules/sinus tracts/scars.	Repeated I&D without chronic plan.	Topical clindamycin for mild; systemic/biologic/surgical escalation.
Atopic dermatitis	Chronic itch, flexural eczema, atopy.	Steroid fear or overuse on face.	Emollients + appropriate topical steroid/calcineurin inhibitor.
Contact dermatitis	Exposure-pattern rash; allergic delayed type IV.	Missing occupational/cosmetic/topical triggers.	Avoid culprit + topical steroids/cool compresses.
Psoriasis	Well-demarcated plaques with silvery scale/nail changes.	Ignoring arthritis or using systemic steroid bursts.	Topicals for mild; phototherapy/systemics/biologics if severe.
Seborrheic dermatitis	Greasy scale in sebaceous areas.	Treating only inflammation without antifungal maintenance.	Ketoconazole/selenium/zinc/ciclopirox +/- brief steroid.
Pityriasis rosea	Herald patch then Christmas-tree trunk eruption.	Confusing with tinea or syphilis in atypical cases.	Reassurance and itch control.
Lichen planus	Pruritic purple polygonal planar papules.	Missing oral/genital/scalp/nail involvement.	Topical steroids; specialist care for erosive/widespread disease.
Urticaria/angioedema	Migratory wheals or deeper swelling.	Using antihistamines instead of epinephrine for anaphylaxis.	H1 antihistamines; airway/epinephrine if systemic.

17. Quick review cases

1. A 19-year-old has open comedones, inflammatory papules, and a few pustules on the cheeks and forehead. No scarring. Start benzoyl peroxide plus topical retinoid; counsel that irritation can occur and improvement takes weeks. Avoid topical antibiotic monotherapy.
2. A 48-year-old reports flushing with wine and hot drinks, central facial erythema, telangiectasia, and gritty eyes. The absence of comedones and presence of triggers favor rosacea; consider ocular treatment and ophthalmology if significant.
3. A 27-year-old smoker has recurrent painful axillary nodules with draining tracts and scars. This is hidradenitis suppurativa, not simple recurrent boils. Stage severity and use chronic anti-inflammatory therapy; counsel smoking cessation.
4. A nurse has chronic fissured hand dermatitis worsened by wet work and detergents. Irritant contact dermatitis is likely; barrier protection, glove strategy, moisturizers, and topical steroids for flares are central.
5. A gardener develops linear vesicles on the forearm two days after clearing brush. Allergic contact dermatitis from poison ivy/oak/sumac is likely; severe widespread disease may require systemic steroid taper long enough to prevent rebound.
6. A patient has extensor plaques with thick silver scale and nail pitting plus morning stiffness in fingers. Treat the skin but also evaluate for psoriatic arthritis and consider systemic therapy referral.
7. A young adult has one oval scaly patch followed by multiple oval trunk lesions along cleavage lines. Pityriasis rosea is likely; if palms/soles or STI risk are present, test for syphilis.
8. A patient on lisinopril develops tongue swelling without hives or itching. Treat airway risk urgently and consider bradykinin-mediated ACE inhibitor angioedema; antihistamines may not work.
9. A child develops hives, wheezing, vomiting, and hypotension after peanuts. This is anaphylaxis; give IM epinephrine first, not antihistamines first.
10. A patient develops fever, facial edema, diffuse rash, eosinophilia, and transaminitis five weeks after starting an anti-seizure medication. This pattern suggests DRESS and needs urgent drug cessation and systemic evaluation.

18. References and source notes

- User-provided dermatology source PDF for project coverage guidance: key definitions; acne vulgaris; rosacea; hidradenitis suppurativa; atopic/contact dermatitis; psoriasis; seborrheic dermatitis; pityriasis rosea; lichen planus; urticaria; angioedema; anaphylaxis; drug reactions.
- American Academy of Dermatology. Updated guidelines for acne management / Guidelines of care for the management of acne vulgaris. 2024.
- American Academy of Dermatology-National Psoriasis Foundation. Psoriasis clinical guidelines and topical/phototherapy/systemic treatment frameworks.
- Merck Manual Professional Edition. Atopic dermatitis, contact dermatitis, angioedema, allergic and atopic disorders, and dermatologic patient evaluation topics.
- World Allergy Organization. Anaphylaxis guidance 2020: intramuscular epinephrine as first-line treatment.